

The Campana–Peternell conjecture in dimension six

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Abstract

We prove that every smooth connected complex projective Fano sixfold with nef tangent bundle is rational homogeneous, resolving the Campana–Peternell conjecture in complex dimension six. As a consequence, a connected compact Kähler manifold X with nef holomorphic tangent bundle and $\dim_{\mathbb{C}} X - \tilde{q}(X) \leq 6$ has ordinary universal cover $F \times \mathbb{C}^{\tilde{q}(X)}$, where F is a rational homogeneous manifold and $\tilde{q}(X)$ is the maximal irregularity of a connected finite étale cover.

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1 Introduction

A smooth connected complex projective variety X is *Fano* if its anticanonical line bundle ω_X^{-1} is ample. A vector bundle E is *nef* if the tautological line bundle $\mathcal{O}_{\mathbb{P}(E)}(1)$ has nonnegative degree on every curve. We use the quotient convention $\mathbb{P}(E) = \text{Proj Sym } E$. A variety is *rational homogeneous* if it is isomorphic to G/P , where G is a connected semisimple complex algebraic group and P is a parabolic subgroup.

The tangent bundle of a rational homogeneous variety is generated by global sections and is therefore nef. Mori’s characterization of projective space by ampleness of its tangent bundle [25] makes the corresponding question for nef tangent bundles natural. The Campana–Petersen conjecture asks whether every smooth complex Fano variety with nef tangent bundle is rational homogeneous [5]. Thus it predicts that a numerical positivity condition determines a variety with a transitive algebraic group action. We prove the conjecture in dimension six.

Theorem 1.1. *Every smooth connected complex projective Fano variety of dimension six with nef tangent bundle is rational homogeneous.*

The Picard number $\rho(X)$ is the rank of the group of divisor classes modulo numerical equivalence, and the *pseudoindex* is the least anticanonical degree of a rational curve on X . These two invariants organize much of the progress on the conjecture. Campana and Petersen established it in dimensions at most three and for fourfolds of Picard number greater than one [5, 6]. The remaining fourfold case was settled through Mok’s and Hwang’s study of the tangent directions of minimal rational curves [22, 16]. Watanabe proved the fivefold case of Picard number greater than one [29, Theorem 1.2], and Kanemitsu completed the fivefold classification [19, Theorem 0.2].

Kanemitsu also proved that a smooth complex Fano n -fold with nef tangent bundle is rational homogeneous whenever $\rho(X) > n - 5$ [18, Theorem 0.2]. For a sixfold, this covers every $\rho(X) \geq 2$, leaving only Picard number one. In that range, standard characterization results reduce the problem to pseudoindices four and five. Watanabe treated the pseudoindex-four sixfold case [30, Theorem 2.4]. We include an independent intersection argument for it; the principal new step is pseudoindex five, where a characteristic-number constraint leads to the Grassmannian.

The difficulty is to pass from positivity of T_X to a specific projective model for the tangent directions of minimal rational curves. Nefness gives strong smoothness and deformation properties, but these properties do not by themselves identify the threefold of directions that occurs in pseudoindex five. We extract additional information by using the universal family in two ways: as a family of rational curves over their parameter space, and as a family of threefolds over X . The compatibility of these two maps yields a characteristic-number identity strong enough to determine the projective variety of tangent directions.

For a covering family of rational curves of least anticanonical degree, the closure of the tangent directions to its members smooth at a general point is called the *variety of minimal rational tangents* (VMRT). Characteristic classes and the geometry of minimal curves already interact in Mok’s work on one-dimensional varieties of minimal rational tangents [22]. Under an additional fourth-Betti-number hypothesis, his argument uses the two rational-curve fibrations of the universal family and characteristic-class inequalities for a rank-two bundle on the parameter space to constrain the tangent image. Hwang removed that hypothesis [16, Section 4]; the resulting classification is stated in [23, Theorem 15]. A related structural approach studies elementary Mori contractions, the contractions of extremal rays: if all of these contractions of a Fano manifold are smooth $\mathbb{P}^1$ -fibrations, the manifold is a complete flag variety [26, Theorem 1.2].

These earlier results show how multiple curve fibrations can constrain both characteristic classes and homogeneous structure. The calculation here concerns a three-dimensional tangent image. It uses the two fibrations to produce explicit linear intersection identities, then combines them with the product identities furnished by the projection formula.

1.1 The argument

Assume $\rho(X) = 1$. Standard characterization theorems leave pseudoindices four and five; the other possibilities are excluded or give projective space $\mathbb{P}^6$ and the smooth quadric Q^6 . The proof is organized around the universal family of rational curves of least degree. Its evaluation map and its projection to the parameter space are both smooth. Along every fibre of the latter map, the relative tangent bundle of evaluation is a sum of copies of $\mathcal{O}_{\mathbb{P}^1}(-1)$.

For pseudoindex four, Kanemitsu's factorization theorem reduces the problem to a variety admitting two fibrations. We rule out the possible fibres by Hard Lefschetz and intersection theory. The last case is an odd-dimensional quadric bundle: its orthogonal description forces a vanishing characteristic number, whereas the second fibration would make an integral divisor have nonintegral degree.

For pseudoindex five and a general point $x \in X$, the evaluation fibre F is a smooth threefold parametrizing minimal rational curves marked at x . Sending a marked curve to its tangent direction at x gives a finite morphism $F \rightarrow \mathbb{P}(T_x^*X) \simeq \mathbb{P}^5$. Its image is the variety of minimal rational tangents, and the pullback L of $\mathcal{O}(1)$ is ample. At this point the degree of the map and the projective type of its image are both undetermined. The central numerical step is

$$32 - 20u + 3u^2 = 0, \quad u = \frac{c_1(T_F) \cdot L^2}{L^3}.$$

We derive this identity from the cohomology of the universal family, Deligne's fixed-part theory [10], and Grothendieck–Riemann–Roch [3]. All resulting characteristic-number constraints are written explicitly. A finite rational calculation then extracts the quadratic relation.

This method separates two useful ingredients. The two fibrations give linear relations among intersection numbers; integration along evaluation also makes selected numbers factor as products. Multiplying these relations forces a nonlinear constraint on the polarized fibre. The proof explains both ingredients and gives an exact certificate for the finite elimination.

Finally, this image spans $\mathbb{P}^5$. The sectional-genus formula excludes $u = 4$ and forces $L^3 = 3$ when $u = 8/3$. The variety of minimal rational tangents is consequently the Segre threefold $\mathbb{P}^1 \times \mathbb{P}^2 \subset \mathbb{P}^5$. The VMRT method of Hwang and Mok [17, Propositions 1.2.1–1.2.2] supplies the nondegeneracy criterion, and Mok's recognition theorem [24] identifies X with $\mathrm{Gr}(2, 5)$, the Grassmannian of two-dimensional subspaces of $\mathbb{C}^5$.

1.2 Consequences of the classification

The classification also has a consequence for compact Kähler manifolds that need not be projective. A differential-geometric antecedent is Mok's uniformization theorem for compact Kähler manifolds with nonnegative holomorphic bisectional curvature [21]. For nef tangent bundles, the structure theorem of Demailly–Peternell–Schneider relates the compact Kähler problem to the Fano conjecture through the fibres of the Albanese map [12, Main Theorem and Theorem 3.14]. To state the consequence precisely, a holomorphic vector bundle is *nef in the analytic sense* when its tautological line bundle admits smooth Hermitian metrics whose curvature is bounded below

by $-\varepsilon\omega$ for every $\varepsilon > 0$, for a fixed Hermitian form ω on its projectivization. This agrees with the preceding numerical definition in the projective setting [12, Section 1.A and Definitions 1.2 and 1.9]. For a connected compact Kähler manifold X with nef tangent bundle, put

$$\tilde{q}(X) = \max\{h^1(X', \mathcal{O}_{X'}) : X' \rightarrow X \text{ is a connected finite étale cover}\}.$$

This maximum exists and is at most $\dim_{\mathbb{C}} X$ by the Demailly–Peternell–Schneider Albanese theorem [12, Proposition 3.9 and Theorem 3.14].

Corollary 1.2 (Albanese fibres and uniformization). *Let X be a connected compact Kähler manifold whose holomorphic tangent bundle is nef in the analytic sense. Suppose*

$$0 \leq d = \dim_{\mathbb{C}} X - \tilde{q}(X) \leq 6.$$

Then there is a connected finite étale cover $X' \rightarrow X$ with the following properties.

1. *The Albanese map $\alpha : X' \rightarrow A = \text{Alb}(X')$ is a smooth proper surjection with connected fibres, where A is a complex torus of dimension $\tilde{q}(X)$. Every fibre is rational homogeneous of dimension d , with a point allowed when $d = 0$.*
2. *There is one such fibre F for which α is a holomorphically locally trivial F -bundle. The ordinary universal cover of X is biholomorphic to*

$$\tilde{X} \simeq F \times \mathbb{C}^{\tilde{q}(X)}.$$

It is therefore biholomorphic to a quasi-projective variety and is holomorphically convex. Its Remmert reduction is the projection

$$F \times \mathbb{C}^{\tilde{q}(X)} \rightarrow \mathbb{C}^{\tilde{q}(X)}.$$

In particular, these conclusions hold whenever $\dim_{\mathbb{C}} X \leq 6$.

The first conclusion applies the Fano classification fibrewise to the Demailly–Peternell–Schneider fibration. The second also uses local deformation rigidity of rational homogeneous manifolds, the Fischer–Grauert theorem, and the universal-cover splitting theorem of Claudon–Hörling–Kollár. The torus A may be nonprojective, and the finite-cover bundle may remain twisted: the product is asserted only for the ordinary universal cover. We prove the corollary after the sixfold theorem in Section 7.

The Fano classification also gives a positive Kähler–Einstein metric on every Fano manifold of positive dimension at most six with nef tangent bundle; Corollary 7.1 records the normalization of this metric.

Organization. Section 2 establishes the common geometry, including the rational-curve fibration calculation in Section 2.1. Section 3 settles pseudoindeces four. Sections 4 and 5 derive the quadratic relation. Section 6 recognizes the Grassmannian, and Section 7 combines the cases with Kanemitsu’s Picard-number theorem and derives Corollary 1.2. Appendix A includes the complete executable certificate for the finite calculation.

2 Minimal rational curves

We now assume that X has Picard number one. The first task is to construct a smooth universal family of rational curves and determine its relative tangent bundle along every member. These facts will provide the common geometric input for the two remaining pseudoindices.

All cohomology groups in the intersection calculations have rational coefficients. A class in H^{2i} has *weight* i . For a smooth morphism a we write $T_a = \ker(da)$ and $-K_a = c_1(T_a)$. Kodaira vanishing gives $H^i(X, \mathcal{O}_X) = 0$ for $i > 0$. The exponential sequence and the Lefschetz (1,1) theorem therefore give

$$H^2(X, \mathbb{Q}) = \mathbb{Q} c_1(T_X). \quad (2.1)$$

We also use that smooth complex Fano manifolds are rationally connected and simply connected; see [7, Theorems 2.11 and 5.3.2].

The *pseudoindex* of X is the positive integer

$$d = \min\{-K_X \cdot B : B \subset X \text{ is an irreducible rational curve}\}.$$

Choose an irreducible component V of the normalized space of rational curves whose members have degree d . Let

$$\begin{array}{ccc} U & \xrightarrow{e} & X \\ \pi \downarrow & & \\ V & & \end{array} \quad (2.2)$$

be its universal family and evaluation map. Thus a fibre of π is the normalization of the corresponding curve, and its map to X is birational onto that curve.

Proposition 2.1. *Let X be a smooth complex Fano manifold of dimension n with nef tangent bundle, and let V be a component of rational curves of minimal anticanonical degree d . Then U and V are smooth projective varieties, $\dim V = n + d - 3$, the map $\pi : U \rightarrow V$ is a smooth $\mathbb{P}^1$ -fibration, and $e : U \rightarrow X$ is smooth and surjective with connected fibres of dimension $d - 2$. For every fibre C of π ,*

$$T_e|_C \simeq \mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus(d-2)}. \quad (2.3)$$

Proof. The properties of U, V, π, e are the minimal-family theorem recalled in [19, Proposition 1.6]; that statement applies in arbitrary dimension. We recall why its hypotheses hold here. For every normalization $\varphi : \mathbb{P}^1 \rightarrow X$, nefness implies a splitting

$$\varphi^*T_X = \bigoplus_{i=1}^n \mathcal{O}(a_i), \quad a_i \geq 0.$$

In particular, each such map is *free*, meaning that its pulled-back tangent bundle is generated by global sections. Minimality of d rules out a degeneration into a sum of rational curves: each nonzero component would already have anticanonical degree at least d . This is the properness condition for the minimal family. Freeness makes the spaces of maps and the evaluation smooth. Finally, the finite map in the Stein factorization of a smooth proper evaluation is étale; simple connectedness of X makes it an isomorphism, giving connected fibres.

It remains to prove (2.3); here it is essential to allow *every* member of V . Fix a fibre $C \simeq \mathbb{P}^1$ and set $\varphi = e|_C$. Let H be the corresponding component of $\mathrm{Hom}_{\mathrm{bir}}(\mathbb{P}^1, X)$. Since all its maps are free, H is smooth, and

$$T_{[\varphi]}H = H^0(\mathbb{P}^1, \varphi^*T_X).$$

The construction of the normalized rational-curve space gives a principal PGL_2 -bundle $H \rightarrow V$ and a compatible map

$$\lambda : \mathbb{P}^1 \times H \longrightarrow U, \quad e \circ \lambda(t, f) = f(t);$$

see [4, Sections 2.1–2.2]. The normalization on the map space introduces no change because H is smooth. Over $[\varphi]$, the map λ identifies $\mathbb{P}^1$ with C .

For $\xi \in T_{[\varphi]}H$, differentiating λ in the constant parameter direction $(0, \xi)$ produces a section of $T_U|_C$. Its image under de is precisely the section of φ^*T_X represented by ξ . Consequently

$$H^0(C, T_U|_C) \longrightarrow H^0(C, \varphi^*T_X) \quad \text{is surjective.} \quad (2.4)$$

The tangent sequence of π , restricted to C , is

$$0 \longrightarrow \mathcal{O}(2) \longrightarrow T_U|_C \longrightarrow \mathcal{O}^{\oplus(n+d-3)} \longrightarrow 0.$$

It gives $h^0(T_U|_C) = n + d$ and $H^1(T_U|_C) = 0$. The nonnegative splitting of φ^*T_X likewise gives $h^0(\varphi^*T_X) = n + d$. Thus the map in (2.4) is an isomorphism. The cohomology sequence of

$$0 \longrightarrow T_e|_C \longrightarrow T_U|_C \longrightarrow \varphi^*T_X \longrightarrow 0$$

now yields $H^0(C, T_e|_C) = H^1(C, T_e|_C) = 0$. Every line-bundle summand of $T_e|_C$ therefore has degree -1 , proving the assertion. No immersion assumption on φ was used. $\square$

We can now isolate the cases that require further work.

Proposition 2.2. *Let X be a smooth complex Fano sixfold with $\rho(X) = 1$ and nef tangent bundle. Then either $X \simeq \mathbb{P}^6$, $X \simeq Q^6$, or its pseudoindex is 4 or 5.*

Proof. For a nonconstant map $\varphi : \mathbb{P}^1 \rightarrow X$, the differential $\mathcal{O}(2) \rightarrow \varphi^*T_X$ is nonzero. At least one summand of the nonnegative splitting of φ^*T_X therefore has degree at least 2, so $d \geq 2$. If $d = 2$, Proposition 2.1 makes e a smooth proper map of relative dimension zero with connected fibres, hence an isomorphism. The map π would then give a $\mathbb{P}^1$ -fibration $X \rightarrow V$ with $\dim V = 5$. The pullback of an ample divisor on V is nonzero and nef but has degree zero on the fibres. This contradicts $\rho(X) = 1$.

For a general point $x \in X$, the *variety of minimal rational tangents* is the closure in $\mathbb{P}(T_x^*X)$ of the tangent lines at x to members of V smooth at x . The marked family $e^{-1}(x)$ has dimension $d - 2$, and the finite tangent-map theorem gives the same dimension for this variety; see [20, Theorem 3.4]. When $d = 3$, it is therefore one-dimensional. The Mok–Hwang classification [23, Theorem 15] gives only $\mathbb{P}^2$, Q^3 , and the five-dimensional homogeneous contact manifold of type G_2 ; see also [19, Section 1]. None of the three resulting manifolds has dimension six.

If $d \geq 6$, every rational curve $B \subset X$ satisfies $-K_X \cdot B \geq \dim X$. The numerical characterization of Dedieu–Höring [9, Theorem 1.3] gives $X \simeq \mathbb{P}^6$ or Q^6 . Thus the only cases left are $d = 4$ and $d = 5$. $\square$

2.1 Cohomology of a rational-curve fibration

The two remaining cases use integration along a smooth $\mathbb{P}^1$ -fibration. The following lemma expresses this operation through an involution and a product rule, without requiring the fibration to be the projectivization of a vector bundle.

Lemma 2.3. *Let $p : B \rightarrow D$ be a smooth projective morphism of smooth connected projective varieties whose fibres are isomorphic to $\mathbb{P}^1$. Set $c = c_1(T_p)$ and $\delta = p^*p_*$. Every class on B is uniquely $a + cb$, with a, b pulled back from D , and c^2 is pulled back from D . Consequently*

$$\overline{a + cb} = a - cb$$

defines a ring involution. On even cohomology,

$$\delta(a + cb) = 2b, \quad \bar{\alpha} = \alpha - c\delta\alpha, \quad (2.5)$$

$$\delta(\alpha\gamma) = (\delta\alpha)\gamma + \bar{\alpha}\delta\gamma. \quad (2.6)$$

For a class h of weight one with fibre degree one, both $2h - c$ and $\alpha - h\delta\alpha$ are pulled back from D . If α has top degree on B , then $\int_B \bar{\alpha} = -\int_B \alpha$.

Proof. The restrictions of 1 and $c/2$ form a basis of the cohomology of every fibre. Leray–Hirsch therefore gives the asserted unique decomposition, and fibre integration gives $p_*c = 2$ and $p_*p^*a = 0$. The higher direct images of $\mathcal{O}_B$ are $p_*\mathcal{O}_B = \mathcal{O}_D$ and $R^j p_*\mathcal{O}_B = 0$ for $j > 0$. Relative Grothendieck–Riemann–Roch [3] consequently gives

$$p_*\mathrm{td}(T_p) = 1.$$

Its component of weight one on D is $p_*c^2/12 = 0$. Writing $c^2 = a + cb$ now gives $2b = 0$, so c^2 is pulled back. This proves that negating c preserves products. The first two identities follow from the decomposition, and multiplication of two such decompositions proves (2.6).

Write $h = a_0 + c/2$. Then $2h - c = 2a_0$, while $\alpha - h\delta\alpha = a - 2a_0b$ for $\alpha = a + cb$. Finally, in top degree the summand pulled back from D vanishes by dimension. The involution negates the remaining summand. $\square$

In particular, if $\bar{h} = \ell - h$ and $\ell = 2h - c$, repeated use of (2.6) gives the polynomial identities

$$\delta(h^a) = \sum_{k=0}^{a-1} h^k (\ell - h)^{a-1-k}, \quad \delta(c^a) = (1 - (-1)^a) c^{a-1} \quad (a \geq 1). \quad (2.7)$$

These formulas are obtained by induction, so do not require division by a cohomology class.

3 Pseudoindex four

Watanabe treats this case in [30, Theorem 2.4]. We give an intersection-theoretic argument using the factorization of the universal family established by Kanemitsu. His theorem gives either rational homogeneity or a factorization of the evaluation map; we exclude the latter possibility. The relative-degree calculation below reproves Kanemitsu’s Lemma 4.3, and the exclusion of projective-plane fibres adapts the Betti-number argument in his proof of Theorem 0.2 [19]. In dimension six, the remaining quadric-bundle case is excluded by the orthogonal-bundle calculation given here.

Let X be a smooth connected complex projective Fano sixfold with nef tangent bundle, Picard number one and pseudoindex four. Let $\pi : U \rightarrow V$ and $e : U \rightarrow X$ be the universal family and evaluation map of a minimal rational component. By Proposition 2.1, we have

$$T_e|_C \simeq \mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2} \quad (C \text{ a } \pi\text{-fibre}).$$

Proposition 3.1. *A smooth complex Fano sixfold with nef tangent bundle, Picard number one and pseudoindex four is rational homogeneous.*

Proof. Kanemitsu's Theorem 3.3 and Proposition 4.1 [19] apply in arbitrary dimension; the latter point is explicit in his Remark 4.2. They show that either X is rational homogeneous, or there is a diagram

$$\begin{array}{ccccc} U & \xrightarrow{f} & W & \xrightarrow{g} & X \\ \pi \downarrow & & \downarrow q & & \\ V & \longrightarrow & Y & & \end{array}$$

with $e = g \circ f$, with f, g, π smooth $\mathbb{P}^1$ -fibrations, and with q a smooth elementary Mori contraction, that is, a Mori contraction of relative Picard number one. We exclude the second possibility. Here $\dim W = 7$. Applied to f and π , [19, Remark 2.3] says that all fibres of q are isomorphic to one member of the list

$$\mathbb{P}^1, \quad \mathbb{P}^2, \quad \mathbb{P}^3, \quad Q^3, \quad Q^5, \quad K(G_2).$$

Here Q^r is a smooth quadric of dimension r , and $K(G_2)$ is the five-dimensional rational homogeneous contact manifold of type G_2 . In particular, $\dim Y \geq 2$.

The two relative degrees. Write

$$H = \frac{c_1(T_X)}{4}, \quad h = g^*H, \quad c = c_1(T_g).$$

The classes h, c form a basis of $H^2(W, \mathbb{Q})$: a smooth $\mathbb{P}^1$ -fibration has the rational Leray–Hirsch decomposition, and $H^2(X, \mathbb{Q}) = \mathbb{Q}H$. These classes are algebraic, so the same basis describes $N^1(W)_{\mathbb{Q}}$, the rational divisor classes modulo numerical equivalence. Fix a π -fibre C . The map $e|_C$ is birational to its image, so $f|_C$ is also birational to its image and $h \cdot f(C) = 1$. Set

$$a = c_1(T_f) \cdot C, \quad b = c \cdot f(C).$$

The relative tangent sequence gives $a + b = -2$. Moreover, $T_f|_C$ is a line subbundle of $\mathcal{O}(-1)^{\oplus 2}$, whence $a \leq -1$. Both a and b are integers.

We show that $b < 0$. The ray contracted by q is generated by $f(C)$. Since $\rho(W) = 2$ and q is elementary, $\rho(Y) = 1$. Thus a positive multiple of $c - bh$ is the pullback of an ample rational class on Y . The sign is positive because $c - bh$ has degree two on a g -fibre, whose numerical class is not in the contracted ray. By Lemma 2.3,

$$g_*c = 2, \quad g_*c^2 = 0,$$

and consequently

$$g_*(c - bh)^2 = -4bH.$$

Intersecting with any complete-intersection curve in X proves $b \leq 0$.

If $b = 0$, the square of c has zero integral over $g^{-1}(D)$ for every smooth curve $D \subset X$. Each g -fibre has positive degree against q^*A , for an ample class A on Y , and hence maps onto a curve under q . Since q is surjective and $\dim Y \geq 2$, two such image curves are distinct. Choose a smooth irreducible complete-intersection curve $D \subset X$ through the corresponding two points of X . The inverse image $g^{-1}(D)$ is a smooth irreducible surface, and its image under q contains both of

those distinct curves. Thus this image has dimension two. It follows that $(q^*A)^2 \cdot [g^{-1}(D)] > 0$, a contradiction. We have proved

$$a = b = -1.$$

Since $-K_f \cdot C = a = -1$, Kanemitsu's Remark 2.3 now reduces the possible fibres of q to $\mathbb{P}^2$, Q^3 , and $K(G_2)$.

The projective-plane fibres. Suppose that q has fibres $\mathbb{P}^2$, so that $\dim Y = 5$. Relative ample classes and their powers give Leray–Hirsch bases for both g and q . Writing $b_i(Z) = \dim H^i(Z, \mathbb{Q})$, we obtain $b_2(Y) = 1$, and Poincaré duality on Y gives

$$\begin{aligned} b_4(X) + 1 &= b_4(W) = b_4(Y) + 2, \\ b_6(X) + b_4(X) &= b_6(W) = 2b_4(Y) + 1. \end{aligned}$$

Hence $b_6(X) = b_4(X) - 1$. This contradicts the injectivity of cup-product by an ample class from $H^4(X, \mathbb{Q})$ to $H^6(X, \mathbb{Q})$, which follows from Hard Lefschetz.

The remaining fibres. We may therefore assume that $\dim Y \leq 4$. Put

$$z = \frac{h+c}{2}, \quad m_i = \int_W h^{7-i} z^i \quad (0 \leq i \leq 7).$$

Since $z \cdot f(C) = 0$, the class z is pulled back from $H^2(Y, \mathbb{Q})$. Thus $z^5 = 0$, and

$$m_0 = m_5 = m_6 = m_7 = 0, \quad m_1 = \int_X H^6 > 0.$$

The rational-cohomology involution of g fixes classes pulled back from X and sends c to $-c$. It sends z to $h - z$ and acts by -1 on top cohomology. For $i = 2, 4, 6$, therefore,

$$\sum_{j=0}^i (-1)^j \binom{i}{j} m_j = -m_i.$$

In order, these identities give

$$m_2 = m_1, \quad m_1 - 2m_3 + m_4 = 0, \quad 9m_1 - 20m_3 + 15m_4 = 0.$$

Consequently

$$m_1 = m_2 = 5m_4, \quad m_3 = 3m_4.$$

In particular $m_4 > 0$, so $\dim Y \geq 4$. Thus $\dim Y = 4$ and every fibre of q is Q^3 .

It remains to show that this quadric fibration is incompatible with g . Since $-K_W = 4h + c$ has degree three on $f(C)$, write

$$-K_q = 3h - \lambda z.$$

The coefficient λ is integral: a g -fibre has respective degrees zero and one against h and z , and $-K_q$ is an integral divisor class. We will determine λ from the moments above using the quadric-bundle identity

$$q_*(-K_q)^4 = 0.$$

To prove this identity, first apply Fischer–Grauert [14]: the smooth projective morphism q , all of whose fibres are Q^3 , is analytically locally trivial. Its structure group is $\text{Aut}(Q^3) = \text{PGO}_5(\mathbb{C})$. In

odd rank this group is isomorphic to SO_5 . Explicitly, fix a nondegenerate symmetric 5×5 matrix J defining the quadric. If A represents a projective automorphism and $A^\top J A = \mu J$, then

$$\tilde{A} = \frac{\mu^2}{\det A} A$$

has determinant one and preserves J . This expression is unchanged when A is multiplied by a scalar; it is the unique special orthogonal lift and respects products. The transition functions thus define a rank-five holomorphic vector bundle E equipped with a nondegenerate symmetric form with values in $\mathcal{O}_Y$ and trivial determinant. GAGA algebraizes this bundle and its form [28, Theorems 2–3 and Propositions 15, 18]. The resulting analytic identification of W with the relative quadric is algebraic as well. Accordingly, in the quotient convention for projectivization, $W \subset \mathbb{P}_Y(E)$ is the quadric of class 2ζ , where $\zeta = c_1(\mathcal{O}_{\mathbb{P}(E)}(1))$. Self-duality identifies the line and quotient descriptions of this quadric. If $p : \mathbb{P}_Y(E) \rightarrow Y$ is the projection, adjunction gives

$$-K_q = 3\zeta|_W, \quad q_*(-K_q)^4 = 2 \cdot 3^4 p_* \zeta^5 = 2 \cdot 3^4 c_1(E) = 0.$$

Now the projection formula and $z^5 = 0$ give

$$0 = \int_W (-K_q)^4 z^3 = 3^4 m_3 - 4 \cdot 3^3 \lambda m_4.$$

Since $m_3 = 3m_4$ and $m_4 > 0$, this forces $\lambda = 9/4$, contrary to integrality. This excludes the factorization case and proves the proposition. $\square$

4 Characteristic-class constraints in pseudoindex five

We now assume that X is a Fano sixfold with nef tangent bundle, $\rho(X) = 1$, and pseudoindex five. Let

$$\pi : U \longrightarrow V, \quad e : U \longrightarrow X$$

be its smooth universal family. Thus $\dim U = 9$, $\dim V = 8$, the fibres of π are projective lines, and e has relative dimension three. The preceding geometric argument gives $T_e|_C \simeq \mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 3}$ for every π -fibre C . Our goal is to constrain the polarization of an evaluation fibre by combining the two fibrations.

Put

$$h_X = \frac{c_1(T_X)}{5}, \quad h = e^* h_X, \quad c = c_1(T_\pi), \quad \ell = 2h - c, \quad E = T_e. \quad (4.1)$$

The class h has degree one on a π -fibre. Lemma 2.3 therefore shows that ℓ comes from V and that $\bar{h} = \ell - h$.

4.1 The polarized evaluation fibre

Lemma 4.1. *For a general point $x \in X$, the fibre $F = e^{-1}(x)$ is a smooth irreducible projective threefold, and the tangent direction at the marking defines a finite morphism*

$$\tau_x : F \longrightarrow \mathbb{P}(T_x^* X).$$

Its image $\mathcal{C}_x$ is the variety of minimal rational tangents of V at x . The line bundle $L = \tau_x^ \mathcal{O}(1)$ is ample and generated by global sections, with $c_1(L) = \ell|_F$.*

Proof. Smoothness, projectivity and connectedness follow from the established properties of e . A smooth connected variety is irreducible. Let $V_x \subset V$ be the reduced locus of curves passing through x . The restriction $\pi|_F: F \rightarrow V_x$ is proper and has finite fibres: each curve has only finitely many points on its normalization mapping to x . It is therefore finite. Kebekus's Theorem 3.3(1) [20], applied with the ample line bundle ω_X^{-1} , says that only finitely many members of V_x can be singular at a general x , and that even these curves are immersed at their markings. Since $\dim V_x = 3$, a general member is smooth at x , so $\pi|_F$ is birational. As F is normal, it is the normalization of V_x . The hypotheses of Kebekus's Theorem 3.4 are satisfied: V is an irreducible covering family and V_x is proper. That theorem gives the finite tangent morphism on F .

With our quotient convention for projectivization, $\mathbb{P}(T_x^*X)$ parametrizes lines in T_xX . The differential of e is nonzero on T_π at every marking over x , and identifies $T_\pi|_F$ with the pullback of the tautological subline $\mathcal{O}(-1) \subset T_xX \otimes \mathcal{O}$. Consequently $c_1(L) = -c|_F = \ell|_F$. Pullback by a finite morphism preserves ampleness, and the pullbacks of hyperplane sections generate L . $\square$

4.2 Classes descending to the parameter space

Define r, s, t to be the elementary symmetric polynomials in the three formal roots obtained by adding h to the Chern roots of E . Equivalently,

$$\begin{aligned} r &= c_1(E) + 3h, \\ s &= c_2(E) + 2hc_1(E) + 3h^2, \\ t &= c_3(E) + hc_2(E) + h^2c_1(E) + h^3. \end{aligned}$$

This definition uses only rational cohomology; it does not assert that h is the first Chern class of a line bundle.

Lemma 4.2. *The classes r, s, t are pulled back from V .*

Proof. There is an actual line bundle

$$\mathcal{A} = e^* \omega_X^{-1} \otimes (T_\pi^\vee)^{\otimes 2}, \quad c_1(\mathcal{A}) = 5h - 2c = h + 2\ell.$$

Its degree on every π -fibre is $5 - 4 = 1$. Thus $E \otimes \mathcal{A}$ restricts to the trivial rank-three bundle on each fibre. Cohomology and base change show that $\mathcal{B} = \pi_*(E \otimes \mathcal{A})$ is locally free of rank three. The evaluation map $\pi^*\mathcal{B} \rightarrow E \otimes \mathcal{A}$ is an isomorphism on every fibre, and hence is an isomorphism. The formal roots defining r, s, t are consequently the roots of $\pi^*\mathcal{B}$ minus 2ℓ . Their elementary symmetric polynomials are pulled back from V . $\square$

For the fibre F and polarization L of Lemma 4.1, the definitions and $h|_F = 0$ give

$$c_1(L) = \ell|_F, \quad r|_F = c_1(T_F). \quad (4.2)$$

We shall constrain the ratio

$$u = \frac{\int_F c_1(T_F) c_1(L)^2}{\int_F c_1(L)^3}. \quad (4.3)$$

Its denominator is positive. The intersection identities below will force a quadratic equation for u . In Section 6, this equation and the sectional-genus formula will determine the degree of L .

4.3 Two direct-image identities

We first record why the Hodge-theoretic direct images appearing in Riemann–Roch contribute no positive-degree Chern character.

Lemma 4.3. *Let $f : Z \rightarrow B$ be a smooth projective morphism between smooth connected complex projective varieties, and suppose that B is simply connected. For all $p, j \geq 0$, the vector bundle $R^j f_* \Omega_{Z/B}^p$ is holomorphically trivial. In particular its positive-degree rational Chern character vanishes.*

Proof. The local systems $R^k f_* \mathbb{Q}$ are constant. The global invariant cycle theorem gives a surjection

$$H^k(Z, \mathbb{Q}) \longrightarrow H^0(B, R^k f_* \mathbb{Q}) \simeq H^k(Z_b, \mathbb{Q})$$

for every $b \in B$; see Deligne [10, Theorem 4.1.1(ii)]. Restriction is a morphism of pure Hodge structures. Therefore each Hodge component of $H^k(Z_b, \mathbb{C})$ is spanned by restrictions of classes of the same type on Z . A class on Z restricts to a flat section of the local system, and a class of fixed Hodge type keeps that type on every fibre. Thus the Hodge decomposition, and hence the Hodge filtration, is constant in the flat trivialization. This is also the fixed-part statement of [10, Corollary 4.1.2]. The relative Hodge–de Rham degeneration identifies $R^j f_* \Omega_{Z/B}^p$ with the degree- p graded bundle of the Hodge filtration on $R^{p+j} f_* \mathbb{C} \otimes \mathcal{O}_B$; see [15, Section 2(a)–(c)]. Since that filtration is constant, each graded bundle is trivial. $\square$

The base X is simply connected, so the lemma applies to e . Relative Grothendieck–Riemann–Roch [3, §7, p. 113] for $\mathcal{O}_U$ and $\Omega_{U/X}^1 = E^\vee$ gives

$$e_* \operatorname{td}_i(E) = 0, \quad e_* [\operatorname{td}(E) \operatorname{ch}(E^\vee)]_i = 0 \quad (i > 3). \quad (4.4)$$

Indeed these are the components of weight $i - 3 > 0$ of

$$\operatorname{ch} \left(\sum_j (-1)^j R^j e_* \mathcal{O}_U \right) \quad \text{and} \quad \operatorname{ch} \left(\sum_j (-1)^j R^j e_* \Omega_{U/X}^1 \right),$$

respectively.

These two choices capture the equations from all relative holomorphic forms. Indeed, for a rank-three bundle and $p = 0, 1$, $\bigwedge^{3-p} E^\vee = \det(E^\vee) \otimes \bigwedge^p E$ and $\operatorname{td}(E^\vee) = e^{-c_1(E)} \operatorname{td}(E)$. Replacing the Chern roots by their negatives therefore gives

$$[\operatorname{td}(E) \operatorname{ch}(\bigwedge^{3-p} E^\vee)]_i = (-1)^i [\operatorname{td}(E) \operatorname{ch}(\bigwedge^p E^\vee)]_i.$$

Thus $\Omega_{U/X}^2$ and $\Omega_{U/X}^3$ repeat, up to sign in each weight, the equations from $\Omega_{U/X}^1$ and $\mathcal{O}_U$.

4.4 Explicit polynomial representatives

We now turn the tangent-sequence and Riemann–Roch identities into linear relations among weight-nine intersection numbers on U . The projection formula will then identify selected normalized numbers as products involving u . Set

$$x_i = e^*(i! \operatorname{ch}_i(T_X)), \quad i \geq 1.$$

Then $x_1 = 5h$ and $x_i = 0$ for $i \geq 7$, by dimension. Work in the polynomial ring

$$R = \mathbb{Q}[h, x_2, x_3, x_4, x_5, x_6, \ell, r, s, t],$$

with respective weights $(1, 2, 3, 4, 5, 6, 1, 1, 2, 3)$. Evaluation of these generators as cohomology classes defines a graded homomorphism $R \rightarrow H^{2*}(U, \mathbb{Q})$.

Our unknowns are the integrals of weight-nine monomials in R . We obtain linear relations among them from three sources: classes descending to the eight-dimensional space V , classes pulled back from the six-dimensional space X and their π -integrals, and the two Riemann–Roch vanishings. The tangent sequences make the π -integrals explicit. Recall that $\delta = \pi^* \pi_*$ lowers weight by one. The recurrences below express these operations as polynomials, before we collect the resulting relations into a matrix.

Let p_j be the Newton power sum of the three formal roots with elementary symmetric functions r, s, t , and put $p_0 = 3$. For complete specificity, if $s_1 = r, s_2 = s, s_3 = t$, Newton's identities read

$$p_i = \sum_{j=1}^{\min(3,i)} (-1)^{j+1} s_j \begin{cases} i, & j = i, \\ p_{i-j}, & j < i. \end{cases} \quad (4.5)$$

Define

$$q_i = \sum_{j=0}^i \binom{i}{j} p_j (-h)^{i-j}. \quad (4.6)$$

These represent $i! \operatorname{ch}_i(E)$, with $q_0 = 3$. The two tangent exact sequences on U give, for every $i \geq 1$,

$$q_i + x_i = c^i + \pi^*(i! \operatorname{ch}_i(T_V)).$$

Applying $\delta = \pi^* \pi_*$ and using (2.7) gives $\delta x_i = D_i$, where

$$D_i = (1 - (-1)^i)(2h - \ell)^{i-1} - \sum_{j=0}^{i-1} \binom{i}{j} p_j (-1)^{i-j} \sum_{k=0}^{i-j-1} h^k (\ell - h)^{i-j-1-k}. \quad (4.7)$$

In particular $D_1 = 5$, consistently with $x_1 = 5h$, and D_i evaluates to zero for $i \geq 7$. This calculation uses the polynomial product rule for δ , so remains valid even when the evaluated classes are zero divisors.

For a monomial m in the six base generators $(h, x_2, x_3, x_4, x_5, x_6)$ define Δm recursively by

$$\Delta 1 = 0, \quad \Delta h = 1, \quad \Delta x_i = D_i \quad (2 \leq i \leq 6), \quad \Delta(gn) = (\Delta g)n + (g - c\Delta g)\Delta n. \quad (4.8)$$

In the last formula, remove the first generator g dividing $m = gn$ in the displayed order. By (2.6), this recursively specified polynomial evaluates to δm . No independence of different formal recursion orders is needed.

Finally, write θ_i for a polynomial representing $\operatorname{td}_i(E)$. Through weight nine, put

$$\theta_0 = 1, \quad \theta_i = \frac{1}{i} \sum_{j=1}^i \beta_j q_j \theta_{i-j}, \quad (\beta_1, \beta_2, \beta_4, \beta_6, \beta_8) = \left(\frac{1}{2}, -\frac{1}{12}, \frac{1}{720}, -\frac{1}{30240}, \frac{1}{1209600} \right), \quad (4.9)$$

with the remaining β_j zero. To obtain this formula, apply the logarithmic derivative to the product of the Todd factors:

$$z \frac{d}{dz} \log \frac{z}{1 - e^{-z}} = 1 - \frac{z}{e^z - 1} = \frac{z}{2} - \frac{z^2}{12} + \frac{z^4}{720} - \frac{z^6}{30240} + \frac{z^8}{1209600} + O(z^{10}).$$

Since $\text{ch}(E^\vee) = \sum_{j \geq 0} (-1)^j q_j / j!$, the second polynomial in (4.4) is represented by

$$\psi_i = \sum_{j=0}^i \frac{(-1)^j}{j!} q_j \theta_{i-j}. \quad (4.10)$$

4.5 The linear constraints and their geometric meaning

Let $\mathcal{B}_k$ be the weight- k monomials in $(h, x_2, x_3, x_4, x_5, x_6)$, let $\mathcal{I}_k$ be those in (ℓ, r, s, t) , and let $\mathcal{T}_k$ be all weight- k monomials in R . A negative subscript means the empty list. Monomials are ordered recursively: for an ordered generator list ending in a variable g of weight a , list g^j times the monomials of weight $k - aj$ in the preceding variables, in increasing order of j .

Form a rational matrix A whose columns are indexed by $\mathcal{T}_9$. Its rows are coefficient vectors of the following polynomials. We group them according to their geometric source.

First, for a base monomial m , both $m - h\delta m$ and δm descend to V by Lemma 2.3, as does ℓ . The classes r, s, t descend by Lemma 4.2. Products of these classes of weight nine vanish because $\dim V = 8$. This gives the rows

$$w(m - h\Delta m), \quad m \in \mathcal{B}_i, \quad w \in \mathcal{I}_{9-i}, \quad 0 \leq i \leq 6; \quad (4.11)$$

$$w\Delta m, \quad m \in \mathcal{B}_i, \quad w \in \mathcal{I}_{10-i}, \quad 0 \leq i \leq 6. \quad (4.12)$$

Second, a base monomial m of weight greater than six vanishes, as does δm . Also $x_i = 0$ for $i \geq 7$, so $D_i = \delta x_i$ evaluates to zero. Their multiples give

$$wm, \quad m \in \mathcal{B}_i, \quad w \in \mathcal{T}_{9-i}, \quad 7 \leq i \leq 10; \quad (4.13)$$

$$w\Delta m, \quad m \in \mathcal{B}_i, \quad w \in \mathcal{T}_{10-i}, \quad 7 \leq i \leq 10; \quad (4.14)$$

$$wD_i, \quad w \in \mathcal{T}_{10-i}, \quad 7 \leq i \leq 10. \quad (4.15)$$

Finally, the Riemann–Roch identities (4.4) and the projection formula give zero integrals for

$$w\theta_i, \quad w\psi_i, \quad w \in \mathcal{B}_{9-i}, \quad 4 \leq i \leq 9. \quad (4.16)$$

Zero rows may be omitted. Let

$$v = \left(\int_U m \right)_{m \in \mathcal{T}_9}.$$

The preceding vanishings prove $Av = 0$. The first five row families come from zero cohomology classes; the final family only requires the stated integrals to vanish.

The constraints are linear in the intersection numbers on U . To recover a useful condition on u , we next use the product structure forced by integration along e .

We use invariant monomials of weights three and four because their pushforwards along e lie in $H^0(X, \mathbb{Q})$ and $H^2(X, \mathbb{Q}) = \mathbb{Q}h_X$, respectively. Each pushforward is determined by one scalar.

These two cohomology groups therefore supply scalar product identities. Write the ordered lists $M = \mathcal{B}_6$ and $G = (\mathcal{I}_3, \mathcal{I}_4)$ explicitly as

$$\begin{aligned} M &= (h^6, h^4 x_2, h^2 x_2^2, x_2^3, h^3 x_3, h x_2 x_3, x_3^2, h^2 x_4, x_2 x_4, h x_5, x_6), \\ G &= (\ell^3, \ell^2 r, \ell r^2, r^3, \ell s, r s, t, \ell^4, \ell^3 r, \ell^2 r^2, \ell r^3, r^4, \ell^2 s, \ell r s, r^2 s, s^2, \ell t, r t). \end{aligned}$$

Indices begin at zero, and w_j denotes the weight of G_j . Put $D = \int_F c_1(L)^3 > 0$ and define

$$\xi_i = \frac{\int_X M_i}{\int_X h_X^6}, \quad e_* G_j = D \eta_j h_X^{w_j-3}. \quad (4.17)$$

The first expression uses the corresponding base classes on X . The second defines a scalar because $w_j - 3$ is zero or one and $H^2(X, \mathbb{Q}) = \mathbb{Q} h_X$. We have $\xi_0 = \eta_0 = 1$ and $\eta_1 = u$.

Use the index set

$$J = \{0, 1, 2, 4, 5, 6, 7, 8, 9, 10, 12, 13, 15, 16, 17\}$$

for the selected invariant monomials. This subcollection suffices for the elimination in Section 5 and reduces its size; no vanishing is asserted for the omitted monomials r^3 , r^4 , and $r^2 s$. Let

$$C_1 = \{(i, (j)) : 0 \leq i \leq 10, j \in J, w_j = 3 \text{ or } h \mid M_i\}.$$

Order C_1 by increasing i and then increasing j . For each such pair the monomial M_i/h^{w_j-3} is defined, and the projection formula gives

$$\int_U G_j \frac{M_i}{h^{w_j-3}} = D \int_X h_X^6 \xi_i \eta_j. \quad (4.18)$$

The common factor on the right is $\int_U h^6 \ell^3 > 0$. We now retain all linear relations that $Av = 0$ induces on these selected coordinates. The following construction expresses them as a matrix N . If K is any matrix whose columns form a rational basis of $\ker A$, let P consist of its rows indexed, with repetitions if necessary, by the monomials on the left of (4.18). Let N be the nonzero rows in the reduced row echelon form of a matrix whose rows form a basis of $\ker(P^T)$. Thus N is determined by the ordered coordinates, independently of the chosen basis K . Then

$$N(\xi_i \eta_j)_{(i,(j)) \in C_1} = 0. \quad (4.19)$$

Indeed $v \in \ker A$, so its selected coordinates lie in the column space of P , and division by the single positive scalar in (4.18) gives the displayed equation. This completes the geometric derivation of the finite constraints.

5 The numerical constraint

The geometric argument has produced the bilinear equations (4.19). We now prove that they allow only two values of u . This section is a finite calculation over $\mathbb{Q}$: its input is the explicitly defined matrix A and its selected coordinates C_1 , and its output is a quadratic equation for η_1 .

Lemma 5.1. *Let A , C_1 and N be the rational matrices and index set defined in Section 4. Suppose that $\xi \in \mathbb{Q}^{11}$ and $\eta \in \mathbb{Q}^{18}$ satisfy*

$$\xi_0 = \eta_0 = 1, \quad N(\xi_i \eta_j)_{(i,(j)) \in C_1} = 0.$$

Then

$$32 - 20\eta_1 + 3\eta_1^2 = 0. \quad (5.1)$$

We give the finite certificate, including the distinction between the modular rank bound and the rational identities needed to finish the proof. The exact recurrences and elimination procedure are implemented in Appendix A.

5.1 Multiplying the constraints

Write $y_{i,j} = \xi_i \eta_j$ for the selected bilinear coordinates. Multiplying each equation of $Ny = 0$ by $\eta_a \eta_b$, with $a, b \in J$, gives linear equations in the cubic products

$$v_{i,\text{sort}(j,a,b)} = \xi_i \eta_j \eta_a \eta_b, \quad (i, (j)) \in C_1.$$

Here sort arranges the three indices in nondecreasing order, identifying equal products. We will impose those linear equations without requiring an arbitrary solution v to have this product form. In the resulting larger solution space, we prove that

$$32v_{0,(0,0,0)} - 20v_{0,(0,0,1)} + 3v_{0,(0,1,1)} = 0. \quad (5.2)$$

For the product vector, $\xi_0 = \eta_0 = 1$ makes the left side $32 - 20\eta_1 + 3\eta_1^2$. Thus a linear statement on cubic coordinates will prove the desired quadratic relation. The following spaces make this relaxation precise; the intermediate quadratic space also makes its computation smaller.

For $n = 2, 3$, form C_n from C_{n-1} by appending an index $j \in J$ to its tuple and sorting. Repeated coordinates are identified. Equivalently, $(i, z) \in C_n$ means that z is a size- n multiset on J , and either $h \mid M_i$ or z contains an index of weight three. Let $\mathcal{K}_1 = \ker N$. Recursively define $\mathcal{K}_n \subseteq \mathbb{Q}^{C_n}$ by requiring, for every $j \in J$, that the slice

$$(v_{i,\text{sort}(z,j)})_{(i,z) \in C_{n-1}} \text{ belong to } \mathcal{K}_{n-1}.$$

Thus $\mathcal{K}_3$ is the kernel of a single rational matrix: for each unordered pair $a, b \in J$, allowing $a = b$, impose N on the slice

$$(v_{i,\text{sort}(j,a,b)})_{(i,(j)) \in C_1}.$$

There are $\binom{16}{2} = 120$ such pairs. In particular,

$$v_{i,z} = \xi_i \prod_{j \in z} \eta_j \quad (5.3)$$

belongs to $\mathcal{K}_3$: every double slice is a scalar multiple of the given vector in $\ker N$.

Seven monomials M_i contain h , and six of the fifteen indices in J have weight three. Therefore

$$|C_n| = 7 \binom{14+n}{n} + 4 \left(\binom{14+n}{n} - \binom{8+n}{n} \right),$$

which gives $|C_1| = 129$, $|C_2| = 1140$ and $|C_3| = 6820$.

5.2 An upper bound from finite-field elimination

The coefficient recurrences in Section 4 give 1785 nonzero rows and 581 columns for A . Exact elimination gives a matrix K of size 581×65 with $AK = 0$. The selected-row matrix P has size 129×65 . Choose N in reduced row echelon form. The following checks give, in particular, complete bases rather than merely lists of relations:

matrix	size	rank modulo 10007
A	1785×581	516
K	581×65	65
P	129×65	60
N	69×129	69

The identities $AK = 0$ and $NP = 0$ hold over $\mathbb{Q}$. Every denominator in these matrices is prime to 10007. Consequently the first two ranks prove that K is a basis of $\ker A$, and the last two prove that N is the full left annihilator of P . Write $\mathcal{K}_n(\mathbb{F}_{10007})$ for the spaces defined by the same slice equations using the reduction of N modulo 10007.

Here is the elimination procedure used for the larger spaces. Suppose that the columns of Z are a basis for the solutions on a union S of slices, and the columns of B are a basis for the solutions on the next slice T . On the overlap, the two parameter vectors a, b must satisfy

$$Z_{S \cap T} a - B_{S \cap T} b = 0.$$

If the columns of $\begin{pmatrix} V_1 \\ V_2 \end{pmatrix}$ form the kernel of this matrix, then the columns of

$$\begin{pmatrix} ZV_1 \\ B_{T \setminus S} V_2 \end{pmatrix}$$

parametrize all solutions on $S \cup T$. This parametrization is injective: zero on the union is zero on each piece, and the old parametrizations were injective. Appending a fixed index to a multiset is itself injective, so there are no hidden identifications within an individual slice. Ordinary Gaussian elimination and successive application of this construction give

$$\dim \mathcal{K}_1(\mathbb{F}_{10007}) = 60, \quad \dim \mathcal{K}_2(\mathbb{F}_{10007}) = 163, \quad \dim \mathcal{K}_3(\mathbb{F}_{10007}) = 6. \quad (5.4)$$

For the rational conclusion, use the single double-slice matrix defining $\mathcal{K}_3$. It has $69 \cdot 120 = 8280$ rows and 6820 columns, and its denominators are units modulo 10007. Its modular rank is 6814, so its rational rank is at least 6814. Thus

$$\dim_{\mathbb{Q}} \mathcal{K}_3 \leq 6. \quad (5.5)$$

This argument does not assume that intermediate rational kernel bases specialize to bases.

5.3 Six rational columns

It remains to exhibit six independent rational solutions and show that the functional giving the desired quadratic vanishes on their span. These columns are witnesses for the linear system: no assertion that they arise from varieties is needed. Three have product form, and three are specified by their small supports. The relaxed system allows both kinds without imposing the original normalization or requiring a product representation. Define the following rational vectors:

$$\begin{aligned} B(v) &= (1, 7 - 2v/5, 45 - 4v, 275 - 30v, 11, 65 - 2v, \\ &\quad 85, 17, 95 - 2v, 25, 35), \\ B' &= (1, 11/5, 5, 11, -1, -1, 10, -3, -1, 0, 14), \\ H^a &= (1, 4, 16, 64, 6, 24, 4, 3, 12, \\ &\quad 48, 192, 768, 17, 68, 272, 96, 9, 36), \\ H^b &= \frac{1}{3}(3, 8, 21, 54, 9, 24, 6, 14, 37, \\ &\quad 97, 252, 648, 37, 97, 252, 96, 16, 42). \end{aligned}$$

For $(B, H) = (B(0), H^a), (B(1), H^a), (B', H^b)$, let

$$v(B, H)_{i,z} = B_i \prod_{j \in z} H_j.$$

Direct rational substitution gives $N(B_i H_j)_{C_1} = 0$, so these three columns lie in $\mathcal{K}_3$. Three further columns s_1, s_2, s_3 have the following entries and are zero elsewhere:

i	z	s_1	s_2	s_3
5	(15, 15, 15)	3	0	0
7	(15, 15, 15)	1	0	0
9	(15, 15, 15)	-5	0	0
7	(10, 10, 15)	0	2	0
7	(10, 10, 10)	0	4	1
9	(10, 10, 10)	0	25	0

Applying N to each of their 120 double slices gives zero over $\mathbb{Q}$. These are exact rational equalities, separate from the finite-field calculation.

The six columns are independent. Indeed, restrict them to the rows

$$(0, (0, 0, 0)), \quad (0, (0, 0, 1)), \quad (1, (0, 0, 0)), \\ (5, (15, 15, 15)), \quad (7, (10, 10, 15)), \quad (7, (10, 10, 10)).$$

The upper-right 3×3 block is zero. The upper-left block has determinant $-8/15$, and the lower-right block is $\begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 4 & 1 \end{pmatrix}$, with determinant 6. The full minor is $-16/5$. Together with (5.5), this proves that the six displayed columns form a rational basis of $\mathcal{K}_3$.

Proof of Lemma 5.1. The functional in (5.2) vanishes on the six basis columns. On a displayed product column its value is $32 - 20H_1 + 3H_1^2$, with $H_1 = 4$ or $8/3$; the sparse columns are zero at all three coordinates. Apply it to (5.3) and use $\xi_0 = \eta_0 = 1$. $\square$

Returning to the evaluation fibre, (4.19) verifies the hypothesis of the lemma and $\eta_1 = u$. Hence

$$(u - 4)(3u - 8) = 0.$$

It remains to identify which value is compatible with the tangent map and to recognize the resulting variety of minimal rational tangents.

6 Recognizing the Grassmannian

We finish the pseudoindex-five case by interpreting the numerical constraint on the fibres of the evaluation map. Throughout this section, X is a smooth complex Fano sixfold with nef tangent bundle, Picard number one and pseudoindex five. We use its minimal rational component V , universal family $\pi: U \rightarrow V$, and evaluation $e: U \rightarrow X$. Thus e is smooth with connected three-dimensional fibres. For a general point $x \in X$, let $F = e^{-1}(x)$ and let L be the tangent-map polarization of Lemma 4.1. Our goal is to recover X from the projective geometry of the tangent directions to its minimal rational curves. For this polarization, Lemma 5.1 gives

$$32 - 20u + 3u^2 = 0, \quad u = \frac{c_1(T_F) \cdot L^2}{L^3}. \quad (6.1)$$

We first show that the tangent image spans $\mathbb{P}^5$. Its degree bound and the sectional-genus formula will then determine L^3 and the degree of the tangent map. Finally, normality of the image will let us use the smoothness of F to select the Segre threefold among the varieties of minimal degree.

Lemma 6.1. *For a general point $x \in X$, the threefold $\mathcal{C}_x \subset \mathbb{P}(T_x^*X) \simeq \mathbb{P}^5$ is linearly nondegenerate.*

Proof. Write $\widehat{\mathcal{C}}_x \subset T_x X$ for its affine cone and $W_x = \text{Span}(\widehat{\mathcal{C}}_x)$. Suppose that W_x is proper. Since $\mathcal{C}_x$ has dimension three, $4 \leq \dim W_x \leq 5$. On a dense open subset of X , these spans define the distribution $W \subset T_X$.

We use two results of Hwang and Mok [17, Propositions 1.2.1 and 1.2.2]. For an irreducible variety of minimal rational tangents on a uniruled projective manifold with $b_2 = 1$, a proper spanned distribution cannot be integrable. On the other hand, it is integrable if the bivectors

$$\alpha \wedge v, \quad \alpha \in \widehat{\mathcal{C}}_x \setminus \{0\} \text{ a smooth point, } v \in T_\alpha \widehat{\mathcal{C}}_x, \quad (6.2)$$

span $\wedge^2 W_x$ at general x . These results apply here: $b_2(X) = 1$ was established earlier, and $\mathcal{C}_x$ is irreducible as the image of F .

We verify the spanning condition directly, without assuming that $\mathcal{C}_x$ is smooth. Otherwise a nonzero alternating form $\omega \in \wedge^2 W_x^*$ annihilates all bivectors (6.2). The one-form $\omega(\alpha, d\alpha)$ vanishes on the smooth cone locus. Its exterior derivative is the restriction of 2ω , so every four-dimensional tangent space $T_\alpha \widehat{\mathcal{C}}_x$ is isotropic. An alternating form of rank $2r$ on an m -dimensional space has isotropic subspaces of dimension at most $m - r$. Since $m \leq 5$ and $r \geq 1$, we must have

$$4 \leq m - r \leq 4,$$

and hence $m = 5$ and $r = 1$.

Projection to the two-dimensional symplectic quotient $W_x / \ker \omega$ has differential of rank at most one on the smooth cone locus. In characteristic zero, the Zariski closure of the image therefore has dimension at most one. It is an irreducible cone, hence is either the origin or a line. But $\widehat{\mathcal{C}}_x$ spans W_x , so its projection spans the two-dimensional quotient, a contradiction. Thus the bivectors do span $\wedge^2 W_x$, making W integrable and contradicting the other Hwang–Mok result. $\square$

Proposition 6.2. *Every smooth complex Fano sixfold with nef tangent bundle, Picard number one and pseudoindex five is isomorphic to $\text{Gr}(2, 5)$.*

Proof. The geometric constraints (4.19) and Lemma 5.1 give $u = 4$ or $u = 8/3$. We first determine the degree of L . By Bertini, two general members of $|L|$ meet in a smooth curve B . Both successive divisors are connected: for a smooth projective variety Z of dimension at least two and an ample divisor D , Kodaira vanishing and Serre duality give $H^1(Z, \mathcal{O}_Z(-D)) = 0$, and the divisor exact sequence then gives $H^0(D, \mathcal{O}_D) = \mathbb{C}$. Applying this first to F and then to its smooth ample surface section proves that B is connected. Adjunction gives

$$g(B) = 1 + \frac{(K_F + 2L) \cdot L^2}{2} = 1 + \left(1 - \frac{u}{2}\right) L^3. \quad (6.3)$$

Here and below L^3 denotes its top intersection number.

By Lemma 6.1, $\mathcal{C}_x$ is a nondegenerate threefold in $\mathbb{P}^5$, so its degree is at least three. The degree formula for the finite morphism τ_x yields

$$L^3 = \deg(\tau_x) \deg(\mathcal{C}_x) \geq 3.$$

If $u = 4$, Equation (6.3) gives $g(B) = 1 - L^3 < 0$. Thus $u = 8/3$, and the same equation gives $g(B) = 1 - L^3/3 \geq 0$. It follows that

$$L^3 = 3, \quad \deg(\tau_x) = 1, \quad \deg(\mathcal{C}_x) = 3.$$

The del Pezzo–Bertini classification of varieties of minimal degree [13, Theorem 0.1] now makes $\mathcal{C}_x$ a rational normal scroll, possibly a cone. Indeed, linear spaces and quadrics have smaller degree, while a cone over the Veronese surface has degree four. A three-dimensional scroll of degree three has parameters $a_1, a_2, a_3 \geq 0$ with $a_1 + a_2 + a_3 = 3$.

To turn the finite degree-one tangent map into an isomorphism, we need normality of its target. These scrolls are normal, including the cone cases, as follows. Their homogeneous coordinate rings are generated by the monomials

$$z_i s_0^j s_1^{a_i-j}, \quad 1 \leq i \leq 3, \quad 0 \leq j \leq a_i.$$

These monomials generate exactly the invariant subring of $\mathbb{C}[z_1, z_2, z_3, s_0, s_1]$ for the multiplicative-group action with weights $(-a_1, -a_2, -a_3, 1, 1)$, graded by total degree in the z_i . In this grading every displayed generator has degree one. To see that they generate the invariant ring, distribute the s_0 and s_1 exponents of an invariant monomial among its z_i factors, each of which requires exactly a_i such exponents. An invariant subring of a normal domain is normal: an element of its fraction field integral over it belongs to the original domain and remains invariant. Thus the scroll is normal.

The finite birational morphism $\tau_x: F \rightarrow \mathcal{C}_x$ is therefore an isomorphism. In particular, $\mathcal{C}_x$ is smooth. If some a_i vanished, the scroll would be a nontrivial cone over a nonlinear variety and hence singular along its vertex. Consequently all a_i are positive, so $(a_1, a_2, a_3) = (1, 1, 1)$. We have proved the projective equivalence

$$\mathcal{C}_x \simeq \mathbb{P}^1 \times \mathbb{P}^2 \hookrightarrow \mathbb{P}^5, \quad \text{embedded by } \mathcal{O}(1, 1).$$

This is the variety of minimal rational tangents of $S = \text{Gr}(2, 5)$. At a two-plane $A \subset \mathbb{C}^5$, $T_A S = \text{Hom}(A, \mathbb{C}^5/A)$, and tangent directions to lines are exactly the rank-one maps. Their projectivization is the displayed Segre variety. For example, in graph coordinates the Plücker coordinates consist of the constant coordinate 1, the entries of a map, and its 2×2 minors; a direction determines a line precisely when the quadratic minors vanish.

Finally, Mok’s recognition theorem [24, Main Theorem, §2.1] states that a Fano manifold of Picard number one whose general variety of minimal rational tangents is projectively equivalent to that of an irreducible compact Hermitian symmetric space is isomorphic to that space. The model $S = \text{Gr}(2, 5)$ has rank two, and we have verified the required projective equivalence for the minimal component V . The theorem gives $X \simeq S$. $\square$

7 The sixfold theorem and its consequences

Proof of Theorem 1.1. Let X satisfy its hypotheses. Since X is projective and positive dimensional, an ample divisor has nonzero numerical class, so $\rho(X) \geq 1$. If $\rho(X) \geq 2$, then $\rho(X) > \dim X - 5$. Kanemitsu’s Theorem 0.2 [18] applies and gives rational homogeneity.

Suppose now that $\rho(X) = 1$. By Proposition 2.2, either X is $\mathbb{P}^6$ or Q^6 , or its pseudoindex is four or five. The first two varieties are rational homogeneous. Proposition 3.1 settles pseudoindex four. In pseudoindex five, the characteristic-class identities of Section 4 and Lemma 5.1 give $32 - 20u + 3u^2 = 0$. Proposition 6.2 therefore gives $X \simeq \text{Gr}(2, 5)$, which is rational homogeneous. These cases exhaust the possibilities. $\square$

7.1 A Kähler–Einstein consequence

Corollary 7.1. *Let X be a smooth connected complex projective Fano variety with nef tangent bundle and $1 \leq \dim_{\mathbb{C}} X \leq 6$. Then X admits a positive Kähler–Einstein metric. More precisely, with the Ricci-form convention*

$$\rho(\omega) = -i\partial\bar{\partial} \log \det(g_{a\bar{b}}),$$

there is a Kähler form ω such that

$$\rho(\omega) = \omega, \quad [\omega] = 2\pi c_1(-K_X).$$

Proof. In dimension six, Theorem 1.1 makes X rational homogeneous. For $1 \leq n = \dim_{\mathbb{C}} X \leq 5$, its Picard number is at least one and hence greater than $n - 5$, so Kanemitsu’s Theorem 0.2 [18] gives the same conclusion. Thus the lower-dimensional cases already follow from the known classification. The homogeneous metric construction of Alekseevskii and Perelman [1], recalled in [11, §1, p. 285], supplies a positive Kähler–Einstein metric on X . Write its Kähler form as σ , with $\rho(\sigma) = \lambda\sigma$ and $\lambda > 0$. Constant rescaling does not change the Ricci form, so $\omega = \lambda\sigma$ satisfies $\rho(\omega) = \omega$. Since the Ricci form represents $2\pi c_1(-K_X)$ in this convention, the asserted class follows. $\square$

7.2 Albanese fibres and the ordinary universal cover

We now combine the classification with the analytic structure theorems for compact Kähler manifolds with nef tangent bundle.

Proof of Corollary 1.2. If $\dim_{\mathbb{C}} X = 0$, connectedness makes X a point and every assertion follows. Assume henceforth that $\dim_{\mathbb{C}} X > 0$. Every finite étale cover remains compact Kähler, and its tangent bundle is analytically nef by Proposition 3.4 of Demailly–Peternell–Schneider. Their Proposition 3.9 [12], applied to every connected finite étale cover, makes its Albanese map surjective; hence its irregularity is at most $\dim_{\mathbb{C}} X$. These irregularities are nonnegative integers and include that of X , so their maximum $q = \tilde{q}(X)$ is attained. Choose a connected finite étale cover $X' \rightarrow X$ attaining it. The Main Theorem and Theorem 3.14 of Demailly–Peternell–Schneider, together with their Proposition 3.9, give a smooth proper Albanese surjection

$$\alpha : X' \longrightarrow A = \text{Alb}(X')$$

with connected fibres, where A is a complex torus of dimension q . Every positive-dimensional fibre is a smooth projective Fano manifold with nef tangent bundle and has dimension $d = \dim_{\mathbb{C}} X - q$. Analytic nefness agrees with numerical nefness on these projective fibres, so the algebraic classification applies.

If $d = 6$, Theorem 1.1 makes every fibre rational homogeneous. For $1 \leq d \leq 5$, a projective fibre F_a has $\rho(F_a) \geq 1 > d - 5$, and Kanemitsu’s Theorem 0.2 [18] gives the same conclusion. This is precisely the established lower-dimensional range recalled in the introduction. If $d = 0$, each connected fibre is a point. This proves the first part of the corollary.

Suppose first that $d > 0$. Every rational homogeneous manifold G/P satisfies

$$H^1(G/P, T_{G/P}) = 0$$

and is locally rigid under complex deformations [2, Introduction and Section 3.7]. Applying this at each fibre of the proper holomorphic submersion α shows that the fibre’s biholomorphism type

is locally constant on A . Since A is connected, all fibres are biholomorphic to one fixed rational homogeneous manifold F . The analytic Fischer–Grauert theorem [14], in the form recalled in [27, Theorem 1.1], now makes α holomorphically locally trivial. If $d = 0$, the proper submersion with one-point fibres is a biholomorphism, so this conclusion also holds with F a point.

The universal cover of the complex torus A is $\mathbb{C}^q$, which is Stein and contractible even when A is not projective. Theorem 4.4 of Claudon–Hörling–Kollár [8] applies to the locally trivial proper fibration α and gives

$$\widetilde{X}' \simeq \widetilde{F} \times \mathbb{C}^q.$$

Here the tildes denote ordinary universal covers. A rational homogeneous manifold F is simply connected: write it using the simply connected semisimple cover of its acting group; the corresponding parabolic subgroup is connected, and the homotopy exact sequence of that homogeneous fibration gives $\pi_1(F) = 0$. This is also true for a point. Thus $\widetilde{F} = F$. The connected finite étale cover $X' \rightarrow X$ has the same ordinary universal cover as X , proving $\widetilde{X} \simeq F \times \mathbb{C}^q$.

Since F is projective, $F \times \mathbb{A}^q$ is quasi-projective and its analytification is $F \times \mathbb{C}^q$. The projection of this product to $\mathbb{C}^q$ is proper, surjective, and has connected compact fibres. For every open set $U \subseteq \mathbb{C}^q$, a holomorphic function on $F \times U$ is constant on each fibre and hence descends holomorphically to U by restriction to $\{f_0\} \times U$ for any fixed $f_0 \in F$. Thus the projection has direct image of its structure sheaf equal to $\mathcal{O}_{\mathbb{C}^q}$. Since the target is Stein, this projection is the Remmert reduction, and the product is holomorphically convex. The argument includes $q = 0$, when the target is a point. $\square$

A Exact arithmetic certificate

The following complete Python program verifies the finite calculations in Section 5. It requires SymPy and NumPy. Run `python3 verification/verify.py` with assertions enabled to perform the checks described below. The monomial order, coefficient recurrences and projection are exactly those defined in Section 4.

The calculation has three logically separate parts. First, exact rational identities $AK = 0$ and $NP = 0$, together with the four modular ranks in Section 5, certify the kernel and annihilator bases. Reduction of a rational entry explicitly inverts its denominator modulo 10007, so a noninvertible denominator would stop the calculation. Second, the slice-merging procedure computes the finite-field spaces and checks their dimensions. The final basis is also tested directly against every double-slice constraint. Third, rational substitution checks the six displayed columns, their nonzero minor $-16/5$, and the final linear functional.

SymPy performs the polynomial and rational operations exactly. The finite-field arrays use signed 64-bit integers. After each reduction, their entries lie between 0 and 10006. The inner dimension of every matrix product is at most 6820, since a basis has no more columns than the number of coordinates of its space. Every accumulated dot product is therefore at most

$$6820 \cdot 10006^2 = 682818645520 < 2^{63} - 1.$$

The products and subtractions in a pivot step have smaller absolute values. Thus no arithmetic operation can wrap around before modular reduction.

```
from functools import lru_cache
from math import comb, factorial, isqrt
from itertools import combinations_with_replacement
import sympy as sy
```

```

from sympy.polys.rings import ring
from sympy.polys.domains import QQ
from sympy.polys.matrices import DomainMatrix
import numpy as np

R,h,x2,x3,x4,x5,x6,l,r,s,t=ring('h,x2,x3,x4,x5,x6,l,r,s,t',QQ)
one=R.one
base=(h,x2,x3,x4,x5,x6)
inv=(l,r,s,t)
gens=base+inv
wt=dict(zip(gens,(1,2,3,4,5,6,1,1,2,3)))
@lru_cache(None)
def mons(n,gs):
    if n==0: return (one,)
    if n<0 or not gs: return ()
    g=gs[-1]; d=wt[g]
    return tuple(g**i*m for i in range(n//d+1) for m in mons(n-d*i,gs[:-1]))

c=2*h-1
hb=1-h
S=(one,r,s,t)
p=[3*one]
for i in range(1,11):
    p.append(sum((-1)**(j+1)*S[j]*(p[i-j] if i!=j else i)
                for j in range(1,min(3,i)+1)))
D=[R.zero]+[(1-(-1)**i)*c**(i-1)-sum(comb(i,j)*p[j]*(-1)**(i-j)
    *sum(h**k*hb**(i-j-1-k) for k in range(i-j)) for j in range(i))
    for i in range(1,11)]
dg=dict(zip(base,[one]+D[2:7]))
@lru_cache(None)
def delta(m):
    if m==one: return R.zero
    ex=next(iter(m)) # m a base monomial
    g=base[next(i for i,a in enumerate(ex) if a)]
    n=m.quo(g)
    return dg[g]*n+(g-c*dg[g])*delta(n)

q=[sum(comb(i,j)*p[j]*(-h)**(i-j) for j in range(i+1)) for i in range(10)]
beta=[0,QQ(1,2),-QQ(1,12),0,QQ(1,720),0,-QQ(1,30240),0,QQ(1,1209600),0]
td=[one]
for i in range(1,10):
    td.append(QQ(1,i)*sum(beta[j]*q[j]*td[i-j] for j in range(1,i+1)))
def rows():
    for i in range(11):
        for m in mons(i,base):
            if i<=6:
                for w in mons(9-i,inv): yield w*(m-h*delta(m))
                for w in mons(10-i,inv): yield w*delta(m)
            else:
                for w in mons(9-i,gens): yield w*m
                for w in mons(10-i,gens): yield w*delta(m)
        if i>=7:
            for w in mons(10-i,gens): yield w*D[i]
        if 4<=i<=9:
            v=sum(QQ((-1)**j,factorial(j))*q[j]*td[i-j] for j in range(i+1))
            for w in mons(9-i,base):
                yield w*td[i]
                yield w*v

```

```

M=mons(6,base)
G=mons(3,inv)+mons(4,inv)
J=[0,1,2,4,5,6,7,8,9,10,12,13,15,16,17]
C=[(i,(j,)) for i in range(len(M)) for j in J if j<=6 or M[i].degree(h)>0]
inds={next(iter(m)):i for i,m in enumerate(mons(9,gens))}
def mat(polys):
    data=[{inds[m]:v for m,v in f.items()} for f in polys if f]
    return DomainMatrix(dict(enumerate(data)),(len(data),len(inds)),QQ)
def make():
    A=mat(rows())
    assert A.shape==(1785,581)
    KD=A.nullspace(divide_last=True).transpose()
    assert (A*KD).to_Matrix().is_zero_matrix
    K=KD.to_Matrix()
    assert ker(red(A.to_Matrix())).shape==(581,65)
    assert ker(red(K)).shape==(65,0)
    P=mat(M[i].quo(h**(0 if j<=6 else 1))*G[j] for i,(j,) in C).to_Matrix()*K
    N=sy.Matrix.hstack(*P.T.nullspace()).T.rref()[0]
    assert ker(red(P)).shape==(65,5)
    assert ker(red(N)).shape==(129,60)
    assert K.shape==(581,65) and N.shape==(69,129) and (N*P).is_zero_matrix
    return N

prime=10007
assert all(prime % k for k in range(2,isqrt(prime)+1))
assert 6820*(prime-1)**2 < 2**63-1
def ker(A): # columns give kernel mod prime
    B=A%prime
    m,n=B.shape
    piv=[]
    for j in range(n):
        i=len(piv)
        nz=np.flatnonzero(B[i:,j])
        if not len(nz): continue
        u=(B[i+nz[0],:] * pow(int(B[i+nz[0],j]),-1,prime))%prime
        B[i+nz[0],:]=B[i,:]+u
        B=(B-B[:,j:j+1]*u)%prime
        B[i,:]=u
        piv.append(j)
        if i+1==m: break
    free=[j for j in range(n) if j not in piv]
    K=np.zeros((n,len(free)),dtype=np.int64)
    K[piv,:]=-B[:len(piv),free]
    K[free,:]=np.eye(len(free),dtype=np.int64)
    return K%prime
def grow(K,C):
    # merge slice restrictions along J
    ix={}
    n=0
    Z=np.empty((0,0),dtype=np.int64)
    for j in J:
        cc=[(i,tuple(sorted(z+(j,)))) for i,z in C]
        overlap=[k for k,a in enumerate(cc) if a in ix]
        extra=[k for k,a in enumerate(cc) if a not in ix]
        A=np.hstack((Z[[ix[cc[k]] for k in overlap],:],-K[overlap,:]))
        V=ker(A); b=Z.shape[1]
        Z=np.vstack((Z@V[:,b:] % prime, K[extra,:]*V[b:,b:] % prime))

```

```

        for k in extra:
            ix[cc[k]]=n
            n+=1
    return Z,sorted(ix,key=ix.get)
def red(M):
    fun=lambda a: int(a.p)*pow(int(a.q),-1,prime)%prime
    return np.array([[fun(v) for v in row] for row in M.tolist()],dtype=np.int64)
def modN(N):
    Nmod=red(N)
    K=ker(Nmod)
    c=C
    for shape in [(1140,163),(6820,6)]:
        K,c=grow(K,c)
        assert K.shape==shape
        assert len(c)==len(set(c))==shape[0]
    lookup={a:i for i,a in enumerate(c)}
    for pair in combinations_with_replacement(J,2):
        rows=[lookup[(i,tuple(sorted(z+pair)))]] for i,z in C]
        assert not np.any((Nmod@K[rows,:]) % prime)

def samples(N):
    G1=(1,4,16,64,6,24,4,3,12,48,192,768,17,68,272,96,9,36)
    G2=tuple(sy.Rational(i,3) for i in
              (3,8,21,54,9,24,6,14,37,97,252,648,37,97,252,96,16,42))
    def X(a): return (1,7-sy.Rational(2,5)*a,45-4*a,275-30*a,11,65-2*a,85,17,95-2*a,25,35)
    X2=(1,sy.Rational(11,5),5,11,-1,-1,10,-3,-1,0,14)
    products=[(X(0),G1),(X(1),G1),(X2,G2)]
    for x,y in products:
        assert not any(N*sy.Matrix([x[i]*y[j] for i,(j,) in C]))
    Z={(5,(15,15,15)): [3,0,0], (7,(15,15,15)): [1,0,0], (9,(15,15,15)): [-5,0,0],
        (7,(10,10,15)): [0,2,0], (7,(10,10,10)): [0,4,1], (9,(10,10,10)): [0,25,0]}
    for k in combinations_with_replacement(J,2):
        assert not any(N*sy.Matrix([Z.get((i,tuple(sorted(z+k))),[0,0,0]) for i,z in C]))
    def six_entries(i,z):
        return [x[i]*sy.prod(y[j] for j in z) for x,y in products]+Z.get((i,z),[0,0,0])
    selected=[(0,(0,0,0)),(0,(0,0,1)),(1,(0,0,0)),
              (5,(15,15,15)),(7,(10,10,15)),(7,(10,10,10))]
    assert sy.Matrix([six_entries(i,z) for i,z in selected]).det()==-sy.Rational(16,5)
    v0=six_entries(0,(0,0,0))
    v1=six_entries(0,(0,0,1))
    v2=six_entries(0,(0,1,1))
    assert all(32*a-20*b+3*c==0 for a,b,c in zip(v0,v1,v2))

if __name__=='__main__':
    N=make(); print("N done", flush=True)
    modN(N); samples(N)
    print("pass")

```

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